

João Rodrigo Faria

Firmware / Embedded Systems Engineer

📅 2003 Apr 05 🌐 jrpfaria 🌐 jrpfaria 📍 Aveiro, Portugal ☎ +351 912 627 603
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🧩 Summary

Firmware engineer with experience on XPON/L2/L3/OLT networking equipment and published IEEE research on FPGA acceleration for ML. M.Sc. in Computer and Telematics Engineering focused on low-level and real-time systems.

📁 Professional Experience

Firmware Engineer, Dellent 📍 2025 Aug – Present | Aveiro, Portugal

Firmware development for XPON with L2, L3 and OLT Switches @AlticeLabs

- Took ownership of a ~**35K LOC** undocumented **C firmware stack**, reverse-engineering and documenting **3 core modules** to close team knowledge gaps and accelerate onboarding
- Eliminated ~**5K lines** of redundant code through systematic **refactoring**, following **DRY** principles
- Integrated Clang formatting, static analysis, and unit tests into the **GitHub Actions CI pipeline** and streamlined compile scripts for better maintainability
- Produced structured technical documentation for previously undocumented firmware components

Researcher, Instituto de Telecomunicações 📍 2024 Feb – 2025 Jul | Aveiro, Portugal

- Published **2 first-author IEEE papers** (NorCAS 2025, ICECS 2025) and a co-authored Springer paper (ICCSA 2025) on **FPGA-based DNN acceleration** and ML classification
- Developed a code generation framework for DNN deployment on FPGAs via Vitis HLS, optimizing latency, resource utilization, and real-time inference at the edge

📄 Publications

Using Automatic Code Generation to Streamline the Real-Time FPGA 2025 Nov

Deployment of DNNs at the Edge 📍, *IEEE ICECS 2025*

Developed an open-source HLS-oriented code generator in C for rapid deployment of custom DNN architectures on FPGAs targeting beyond 5G scenarios.

FPGA Acceleration of CNNs at the Edge: A Comparative Study on 2025 Oct

High-Level Synthesis Frameworks 📍, *IEEE NorCAS 2025*

Compared HDL Coder and Vitis HLS across latency, resource usage, and development effort for real-time edge ML deployment on FPGA.

Understanding Signal Feature Impact in CNN-Based Modulation 2025 Jul

Classification with Realistic Datasets 📍, *ICCSA 2025*

🎓 Education

M.Sc. in Computer and Telematics Engineering, 2023 Sep – 2025 Jul | Aveiro, Portugal

Universidade de Aveiro 📍

- **Thesis:** Code generation framework for DNN deployment on FPGAs via High-Level Synthesis
- Implemented DNN operations in C for single-cycle II in Vitis HLS, including transcendental activation functions
- Built a parallel code generator that translates GUI input into HLS-ready C files
- Designed sample-by-sample inference architecture with local memory reuse, enabling beyond-5G real-time scenarios

B.Sc. Computer and Telematics Engineering, Universidade de Aveiro 📍 2020 Oct – 2023 Jul | Aveiro, Portugal

🧠 Skills

Languages

C, C++, Java, Python, Bash, VHDL, MIPS Assembly

Hardware & Architecture

Computer Architecture, Digital Systems, FPGA, Vitis HLS, RISC-V

Performance

Concurrent and Object Oriented Programming, Memory Management

CI & Tooling:

Git, GitHub Actions, Clang

Embedded & Real-Time Systems

FreeRTOS, Zephyr RTOS, Linux, Peripheral Inspection

Parallel Computing

CUDA, OpenMP

📁 Projects

CSV-Parser 📍, *A Fully Deterministic, High-Performance CSV Parser in C* 2026 Apr

- Architected 6 parser variants (computed-goto FSM for branchless processors, branch-based, LUT-based) × (scalar, SIMD) as a performance case study in control-flow and memory strategies
- Used per-worker arena allocators for zero-allocation hot-path memory management

🌐 Languages

Portuguese (C2) | English (C2)